Wheeled Mobile Robots

Kinematic Modelling

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Introduction

To the best of my knowledge, the first general study of the kinematics of wheeled mobile robots is the work of Patrick Muir and Charles Neuman [Muir and Neuman, 1986], which is available on the web. The present document is based on the slides of the lectures Wisama Khalil used to give in the Modelling and Control part of the Mobile Robots course. He himself used the book "Theory of robot control" [Canudas-de Witt et al., 1996] as a basis, a source to which the present document remains strongly related. That said, notations are close but different, so using the present document and the book simultaneously can be confusing.

Of course, as a student of the Mobile Robots class, you can decide to study any book or online source about Mobile Robots. But I want to make clear that the material I provide for my part of the course is sufficient to get full mark at the exam. In practice, the most efficient way to study is to use the provided material and solve the suggested exercises rather than trying to sort out the abundance of material available.

The aim of this part of the Mobile Robots course is to explain how to calculate the kinematic model of a certain class of wheeled mobile robots. The reason why we study the kinematic model is that you cannot do much without it. Roughly, the kinematic model will give you the relations between the wheel variables (rotation and orientation) and the motion of the robot body. It helps you answer a fundamental question: "if I want to move the robot at a given velocity and rotation speed, what should be the speed of the wheels?"

But such a question can only be dealt with in a second phase. You already have some experience of robot manipulators, so you know that a 6 DOF (degrees of freedom) robot can move its tool at any velocity and rotation speed from (nearly) any configuration, subject of course to speed limitations. The situation is not the same for a mobile robot. Even if you do not drive, you know that a car cannot move sideways (I would love that: I am terrible at parking maneuvers). Likewise, most mobile robots do not have full mobility: their mobility is limited by kinematic constraints.

In the first part of this course, we will study the kinematic constraints which are generated by the contact between the wheels and the ground. They will allow us to define and calculate the *degree of mobility* of a given robot, which is roughly the number of components of the robot's speed of motion that can be independently set. Then only shall we be able to address the relation between these components of the motion and the speed of the wheels, *i.e.* the *kinematic model*.

Finally, we will address another important practical problem: having calculated which components of the robot motion can be independently set, what joints of the robot (degrees of freedom of the wheels) should be motorized to be able to produce the desired motion?

Chapter 1

Degree of Mobility

Contents of this chapter.

As stated in the introduction, the main goal of this chapter is to find out how the constraints generated by the contact between the wheels and the ground influence robot mobility. This has been formalized in [Canudas-de Witt et al., 1996], in which the degree of mobility is introduced.

We will study the mobility for a given class of mobile robots, and under the assumption of rolling contact, which we will precisely define later.

The reason we assume rolling contact is because rolling contact is *desirable*, as it is energy efficient and smooth. Hence, it is a reasonable and very common design goal. As a consequence, many wheeled mobile robots can be satisfactorily modeled under this assumption, and the resulting models are very useful in a wide range of situations, even though the assumption is not perfectly satisfied. The fact that it leads to fairly simple models does not hurt either.

Other than this assumption, the generality of our course will mainly be limited by the types of wheels we are going to take into account. We will leave behind some non standard wheels. Still, the so-called *conventional wheels* we take into account here are used by a vast majority of mobile robots.

1.1 Definitions and assumptions

Vocabulary

Definition 1 (Speed)**.** *The time derivative of a* ***scalar*** *position variable.*
If the position variable is an ***angle****, the speed is usually called* ***rotation speed****, but the qualifier is often omitted if there is no ambiguity.*

Definition 2 (Velocity)**.** *A* ***vector****, time derivative of the position of a* ***point****.*

Remember: speed is a scalar, velocity is a vector.

In our context, velocities will be either 2-vectors or 3-vectors.

Definition 3 (Twist)**.** *A twist represents the instantaneous motion of a rigid body as an angular speed around an axis and a linear speed along that axis.*

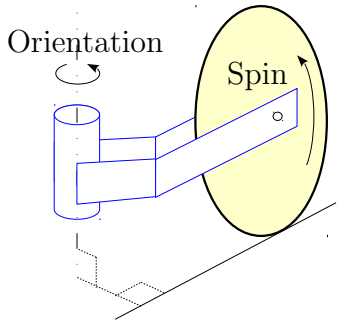


Figure 1.1: Wheel assembly

All points have the same speed along the axis of the twist. In the plane perpendicular to the twist axis, the velocity of a point depends on its position relative to the axis: it is the null vector for a point on the axis and increases in norm when the point gets farther from the axis. In $3D$, the twist has six independent parameters: the four coordinates of the axis, the speed along the axis and the rotation speed around the axis. Equivalently, the three speed components of a given point and three rotation speeds can be used. In our context, the twist will only apply to the rigid platform of the robot (cf. *infra*), which moves in a plane, so we will represent it as a 3-vector with two speed components and one rotation speed component, typically $[\dot{x}, \dot{y}, \dot{\theta}]^T$: we simply suppress the components of the $3D$ motion which are constantly null.

Strictly speaking, the "speed of a solid" is improper. So is the "velocity of a solid" since, except in the case of a pure translation, different points of the solid have different velocities. I will try to resist the temptation, but it is likely the term "speed" will be used here and there to denote a velocity or even a twist since in layman's terms, "speed" encompasses all of the above.

Work assumptions

Definition 4 (Wheeled Mobile Robot)**.** *A wheeled mobile robot is a robot capable of locomotion on a surface solely through the actuation of wheel assemblies mounted on the robot and in contact with the surface [Muir and Neuman, 1986].*

Definition 5 (Wheel assembly)**.** *A wheel assembly is a mechanical device which allows the wheel to rotate around its spin axis. In addition, the device may also allow the wheel to rotate around one orientation axis perpendicular to the spin axis and to the ground (figure 1.1).*

The most general form of wheel assembly taken into account in this course is shown in figure 1.1. In the sequel, wheel assemblies may sometimes be called wheels for convenience.

Assumptions about the **environment of evolution** of the robot:

- The robot moves on a flat horizontal[1] plane.
- The plane is a rigid, non deformable body.

[1] Actually, the fact that the plane is horizontal is not important here in a kinematic modeling part. It would be fundamental if dynamics were considered.

Assumptions about the **robot** itself:

- The robot is a mechanical system consisting of a single rigid body called the *platform*, to which the wheel assemblies are attached.
- The robot has at least three wheel assemblies. To provide stable support on the ground, there should always be at least three wheels whose contact points with the ground are not aligned.

Note that this is more restrictive than the definition of Muir and Neuman, which allows articulated chassis and trailers, when they are excluded here.

Assumptions about **wheels and wheel-ground contact**:

- The wheels are rigid disks (no deformation) with an equal radius[2] r.
- They are perpendicular to the rolling plane.
- There is a single point of contact between a wheel and the rolling plane[3].
- The wheels are always in contact with the ground.
- The motion of the wheels is a *pure rolling motion*, which means that the point of the wheel instantaneously in contact with the ground has zero velocity (figure 1.2.):
  - No slipping: the *tangential* speed v_t (component of the velocity in the plane of the wheel) is zero.
  - No skidding: the *normal* speed v_n (component of the velocity normal to the plane of the wheel) is zero.
  - *Rotational slipping* is allowed, meaning that ω_z can be non zero. (In fact, ω_z has to be non zero when the wheel navigates a turn.)

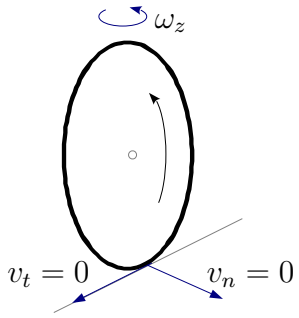


Figure 1.2: Ideal wheel-ground contact.

In this part of the course we are in a purely kinematic context, but let us just recall that pure rolling contact implies that friction forces between the wheel and the ground remain sufficient at all times to prevent slipping and skidding. For example, if a large spinning torque is applied to a wheel, high friction is necessary for the wheel not to slip. Also, if the robot is navigating a turn, large centrifugal forces may be applied perpendicular to the wheel plane, which friction forces will have to balance. With rubber tires, increasing the contact forces can be obtained by using soft rubber (with the downside that it wears

[2] We will show later that this is essentially a notational convenience and does not cause loss of generality.
[3] This is actually a consequence of earlier assumptions.



Figure 1.3: Sweedish (left) and omnidirectional (right) wheels.

out fast) or by increasing the area of contact with the ground (wide tires), in which case pure rolling is likely to become a rough approximation.

The pure rolling contact assumption as defined here excludes some types of wheels, like the so-called *omnidirectional* wheels and *Swedish* wheels. An ominidirectional wheel is such that it can freely skid (move laterally). The model of figure 1.3 (right), is in fact made of a number of small rollers whose spin axes are in the plane of the wheel, thus allowing free skid. The swedish wheel of figure 1.3 (left) is made of small inclined rollers and allows the center of the wheel to move sideways at an angle with respect to the plane of the wheel. Those wheels require careful design in order for *the individual rollers* (not the overall wheel considered as a disk) to operate in conditions close to rolling contact. The design problem is to find a proper shape and placement of the rollers. Again, the design efforts aim at rolling contact of the small rollers, because rolling contact is smooth and energy efficient.

Remark. *Although we do not cover Swedish wheels in this course, I encourage you get a quick look at an on-line video showing an omnidirectional robot built using Swedish wheels. You will very easily find such a video on the web. Knowing that these wheels exist and what can be done with them is part of the expected culture in mobile robots.*

1.1.1 Frames, coordinates and velocity components

We define a frame $R_0(O, x_0, y_0, z_0)$ attached to the plane on which the robot moves, with z_0 perpendicular to the plane and directed upward. The position of the robot will be given with respect to R_0. More precisely, we consider the position and orientation of frame $R_m(M, x_m, y_m, z_m)$ attached to the robot platform, with respect to R_0, as shown on figure 1.4. The vector composed of the position and orientation of the robot is called the *posture* and denoted by ${}^0\xi$. As usual, the left exponent refers to the frame in which the posture is expressed. But whenever ξ is used without left exponent, it means ${}^0\xi$.

$$\xi = {}^0\xi = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \tag{1.1}$$

The orientation θ of the robot is often called *heading*, especially in the context of localization.

There is also a frame associated to each wheel. The wheels are numbered, starting at index 1 and the frame associated to the ith wheel is denoted as R_i. There is no ambiguity,

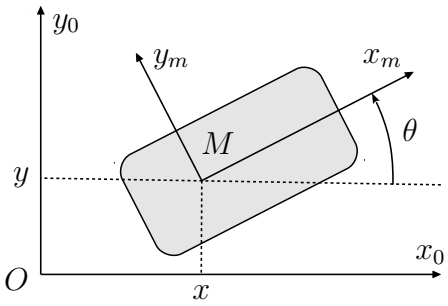


Figure 1.4: The posture of a mobile robot.

since the absolute reference frame is R_0 and wheel frames start at R_1. Figure 1.5 shows the direct frame R_i associated to a wheel, the origin B_i of which is the center of the wheel. The frame R_i is *not attached to the wheel itself* but tho the supporting body of the wheel, represented with dotted lines on figure 1.5. Axis z_i of the frame remains vertical directed upward. Axis y_i is perpendicular to the plane of the wheel, hence the wheel spin φ_i happens around y_i. The point of the wheel in contact with the ground is called C_i. Note that the z axes of all frames are identical: $z_0 = z_m = z_{i,i\geq 1}$.

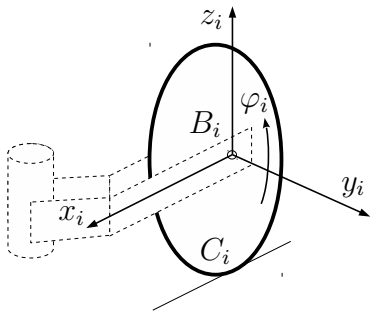


Figure 1.5: Frame and notations related to a wheel.

The twist (velocity and rotation speed) of the platform with respect to frame R_0, sometimes incorrectly referred to as "speed of the robot", is the time derivative of the posture ξ: ${}^0\dot{\xi} = [\dot{x}, \dot{y}, \dot{\theta}]^T$. As far as the rotational speed $\dot{\theta}$ of the robot is concerned, it is always around the z axis common to all frames, so it does not depend on the frame chosen to express it. But the components of the velocity of point M do vary depending on the frame. We will need to express the twist of the robot either in R_0 or in R_m. There is a simple way to change the frame in which $\dot{\xi}$ is expressed, using the following relation:

$$ {}^0\dot{\xi} = {}^0\Omega_m \, {}^m\dot{\xi} \,, \quad with: \;\; {}^0\Omega_m = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.2} $$

Remark. *Care has been taken to use a specific notation for the frame changing matrix applied to the twist, with the idea to avoid confusion. Although equal to such a matrix,* Ω *is* not *an homogeneous transformation of* $\Re^2$*. The notation* Ω *will be used only when changing the frame in which the* twist *is expressed. You can consider it as a notational convenience to represent simultaneously a rotation matrix of* $\Re^2$ *applied to the first two components of* $\dot{\xi}$ *and the identity transformation for the third component, the rotation speed. So equation 1.2 can be seen as a shorthand for:*

$$\begin{bmatrix} {}^0\dot{x} \\ {}^0\dot{y} \end{bmatrix} = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix} \begin{bmatrix} {}^m\dot{x} \\ {}^m\dot{y} \end{bmatrix}$$

and:

$${}^m\dot{\theta} = {}^0\dot{\theta}$$

The first of these two relations only makes sense for vectors *(as the 2D velocity),* not for points*. To change the frame in which a point is expressed, a* $3*3$ *homogeneous matrix must be used (see [Khalil and Dombre, 2004] or your Modeling of Manipulators course). Again, equation 1.2* only makes sense for twist (posture speed) $\dot{\xi}$, not for posture.

Exercise 1. *Write the expression of* ${}^m\dot{\xi}$ *as a function of* ${}^0\dot{\xi}$*.*

1.2 The different types of wheels

We are now going to describe the different types of wheels which we take into account in this course. They are the most commonly used types of wheels in the design of mobile robots, and thus cover a vast majority of actual robots. All these wheels derive from the wheel assembly presented in section 1.1.

The main goal of this section is, for each type of wheel, to do the following:

- Give a schematic representation of the wheel for use in the schematic representation of a robot.
- Determine a set of parameters to define the wheel geometry.
- Determine a set of parameters to define the location of the wheel frame with respect to the robot frame R_m.

The parameters are first presented for the general form of the wheel assembly, then for the particular cases corresponding to the so-called *conventional wheels*.

Our parameters are somewhat different from the ones in [Canudas-de Witt et al., 1996]. The modifications aim at having all wheels derive from a general model. Also, if a given parameter name corresponds to a constant for a given type of robot, it is a constant for all types, contrary to the parameters used in [Canudas-de Witt et al., 1996].

1.2.1 The general wheel assembly

The wheel assembly has been defined in section 1.1. Figure 1.6 gives its parameterization. Parameters L and α locate the orientation axis of the wheel assembly with respect to the platform frame in polar coordinates[4]. Parameters d, β and γ in turn are necessary to locate the wheel frame R_i with respect to the platform. The only variable parameters are β and the spin angle of the wheel φ. Care must be taken that all angles are signed. For α, β and γ, which are rotations around z, positive direction is x to y. For φ, which is a rotation around y_i, positive direction is from z_i to x_i.

The schematic representation and the parameterization uses a top view of the wheel (in $x-y$ plane). The dotted line at the wheel center represents the spin axis y_i.

[4]Polar coordinates are convenient because the angle between $\vec{MA_i}$ and x_m is used in several equations.

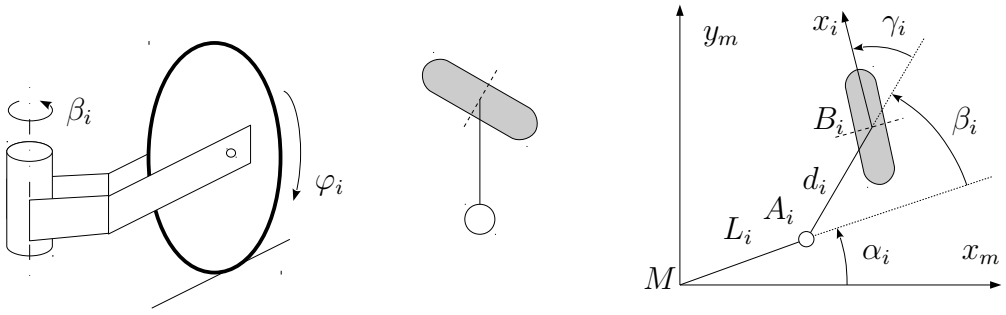


Figure 1.6: General wheel assembly: $3D$ view , symbol and parameterization.

The *conventional wheels* are obtained by setting to zero certain parameters of the general wheel assembly.

1.2.2 The fixed wheel

A fixed wheel is such that its spin axis y_i is fixed with respect to the mobile robot frame R_m. It is completely defined by the position of its point B_i in R_m and the orientation of x_i (figure 1.7). The fixed wheel corresponds to the general wheel assembly, where $A_i = B_i$, $d_i = 0$ and $\beta_i = 0$.

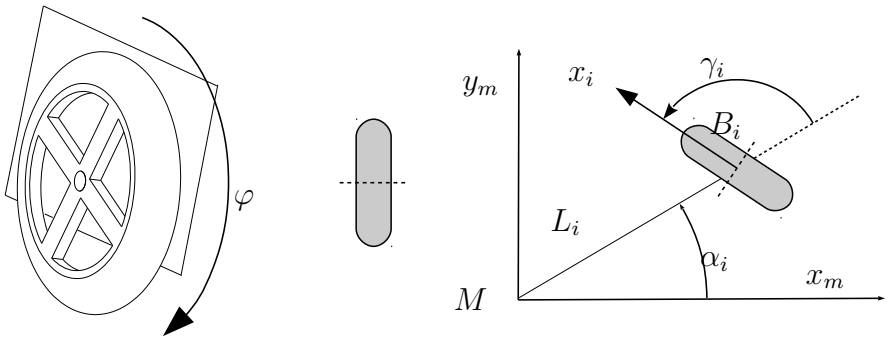


Figure 1.7: Fixed wheel: $3D$ view, symbol and parameterization.

1.2.3 The steerable wheel

A steerable wheel is such that its orientation axis goes through the point of contact of the wheel with the ground (figure 1.8). It corresponds to the particular case of the wheel assembly where $A_i = B_i$, $d_i = 0$ and $\gamma_i = 0$.

Beware of the similarity with the fixed wheel. The symbol of the steerable wheel includes a circle to indicate the presence of the orientation axis. Pay attention to the difference in parameterization: instead of the *constant* γ angle of a fixed wheel, we have a *variable* angle β. The value of β in the schematic representation has no importance, when it should (at least visually) correspond to the real value in the case of a fixed wheel.

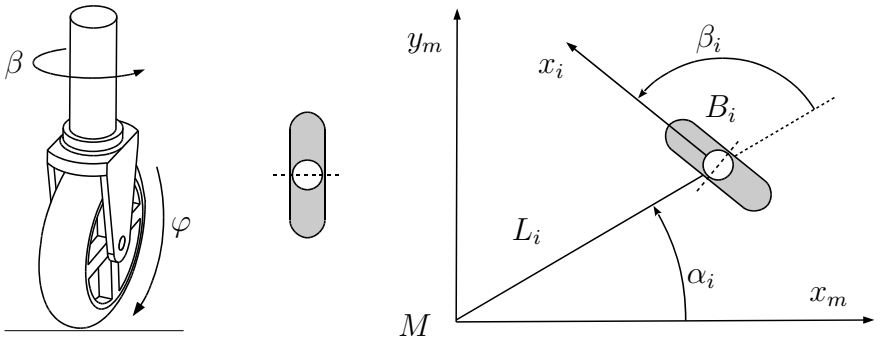


Figure 1.8: Steerable wheel: $3D$ view, symbol and parameterization.

1.2.4 The castor wheel

The castor wheel is such that its orientation axis belongs to the plane of the wheel (figure 1.9). It corresponds to the particular case of wheel assembly where $\gamma = 0$ (conventionally x_i points away from A). By definition, the parameter d of a castor wheel is not equal to zero.

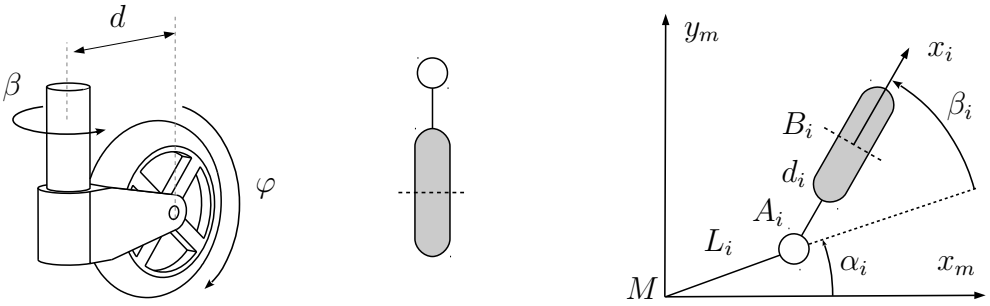


Figure 1.9: Castor wheel: $3D$ view, symbol and parameterization.

1.3 Parameterization of a robot

The parameterization of a robot consists of the following steps:

1. Draw wheel symbols according to robot geometry. Set the robot frame, as much as possible in a way that simplifies the parameters, and indicate characteristic dimensions.
2. Number the wheels in the conventional order: fixed wheels (append suffix "f" to the wheel number), steerable wheels (append "s"), castor wheels (append "c") and general wheels if any (append "g").
3. Draw the x axis of the wheels. For castor wheels, point axis x away from A, so that $\gamma = 0$.
4. Fill the parameter table as shown at table 1.1.

Remark. *The column φ simply reminds the names of the spin angle variables. Same for column β for non fixed wheels. And of course column ψ is redundant. But since angle ψ appears in several equations, it is convenient to write its value once and for all rather than re-evaluate it each time it is required.*

Table 1.1: The parameter table.

Wheel	L	α	d	β	γ	φ	$\psi = \alpha + \beta + \gamma$
$1f$	...	...					
$2\ldots$	...	...					

Exercise 2. *Parameterize the robot of figure 1.10. The wheel configuration does not make sense: it's merely a parameterization exercise which involves all types of wheels.*

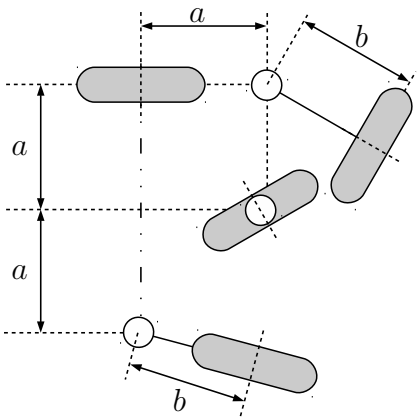


Figure 1.10: Parameterization exercise.

Exercise 3. *Create your own wheel configurations to practice parameterization. As the previous exercise suggests, the wheel configuration need not make sense. You can of course practice on the various types of mobile robots defined later in the course. I suggest you practice in pairs (or in larger groups). Agree on the wheel configuration, wheel numbering, definition of R_m and orientation of x_i axes ; you should then all get the same parameter table. Separately fill the table, then compare the results and settle any disagreement.*

1.4 Pure rolling constraints

We are now going to write the equations corresponding to pure rolling constraints. In order to do this, we need to write the expression of the velocity of point C_i of the wheel in contact with the ground. The constraints will then be expressed by setting the two components of the velocity, expressed in the wheel frame, to zero.

1.4.1 Velocity of the contact point

In this section, we write the equations of the velocity of the contact point for the case of the general wheel, the other types of wheels being simply particular cases of the general wheel. To understand the equations, it is useful to look at the parameterization of the wheel, as defined in figure 1.6.

The velocity of point C is due to:

- The twist (velocity and rotation speed) of the robot platform.
- The orientation speed of the wheel $\dot{\beta}$.
- The spinning speed of the wheel $\dot{\varphi}$.

Other geometric parameters are constant, and hence do not contribute to the velocity of C. As we are considering a single wheel, we made notations lighter by dropping the wheel index where there was no ambiguity. The equations are of course valid for any wheel of the robot. In vector form, we get:

$$\vec{V_c} = \vec{V_M} + \dot{\theta}\vec{z_m} \wedge \vec{MC} + \dot{\beta}\vec{z_m} \wedge \vec{AC} + \dot{\varphi}\vec{y_i} \wedge \vec{BC} \tag{1.3}$$

Each vector involved in the expression is a $3D$ vector. But each term of the sum is a *horizontal* vector, so the velocity of C is horizontal.

We are now going to express the coordinates of $\vec{V_c}$ in the frame of the wheel, R_i. Indeed, in R_i the components of the velocity are the tangential and normal speeds of figure 1.2. The term $\dot{\varphi}\vec{y_i} \wedge \vec{BC}$ is easily expressed in R_i. The other terms are easier to express in R_m and are then projected onto R_i by left-multiplying by the rotation matrix ${}^iR_m$. Now that we work in terms of vector coordinates, we drop the vector arrows and indicate the frame onto which the vector is projected by a left exponent. We get:

$${}^iV_C = {}^iR_m[{}^mR_0\,{}^0V_M + \dot{\theta}z_m \wedge {}^mMC + \dot{\beta}z_m \wedge {}^mAC] + \dot{\varphi}y_i \wedge {}^iBC \tag{1.4}$$

With:

$${}^0V_M = \begin{bmatrix} \dot{x} \\ \dot{y} \\ 0 \end{bmatrix} \quad {}^mAC = \begin{bmatrix} d \cdot cos(\alpha+\beta) \\ d \cdot sin(\alpha+\beta) \\ -r \end{bmatrix} \quad {}^iBC = \begin{bmatrix} 0 \\ 0 \\ -r \end{bmatrix}$$

This is were it is convenient to assume all wheels have the same radius r. With this hypothesis, all points A_i and B_i lie in the same horizontal plane as M. Otherwise, we would have different vertical offsets for the wheels. Should a robot not satisfy the assumption, it is very easy to adapt.

$${}^mMC = {}^mMA + {}^mAC = \begin{bmatrix} L \cdot cos\alpha \\ L \cdot sin\alpha \\ 0 \end{bmatrix} + \begin{bmatrix} d \cdot cos(\alpha+\beta) \\ d \cdot sin(\alpha+\beta) \\ -r \end{bmatrix}$$

And: ${}^iR_m = {}^mR_i^{-1} = rot(z_m, -(\alpha+\beta+\gamma)) = rot(z_m, -\psi) = \begin{bmatrix} cos\psi & sin\psi & 0 \\ -sin\psi & cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

By factoring the time derivatives, namely $\dot{\xi}$, $\dot{\beta}$ and $\dot{\varphi}$, and keeping only the first two components, we obtain the following equations:

$$\begin{bmatrix} v_t \\ v_n \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & d \cdot \sin\gamma + L \cdot \sin(\beta+\gamma) \\ -\sin\psi & \cos\psi & d \cdot \cos\gamma + L \cdot \cos(\beta+\gamma) \end{bmatrix} {}^m\mathbf{\Omega}_0\,\dot{\xi} + \begin{bmatrix} d \cdot \sin\gamma \\ d \cdot \cos\gamma \end{bmatrix} \dot{\beta} + \begin{bmatrix} -r \\ 0 \end{bmatrix} \dot{\varphi} \tag{1.5}$$

With:

$${}^m\Omega_0 = \begin{bmatrix} cos\theta & sin\theta & 0 \\ -sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.6}$$

The expression of ${}^m\Omega_0$ given here is the inverse of the matrix ${}^0\Omega_m$ of section 1.1.1, equation 1.6. Hence, it is also the solution to exercise 1 :-)

We are now going to write the particular forms of system 1.5 for each type of wheel.

1.4.2 Constraints for fixed wheels

The pure rolling constraints correspond to $v_t = 0$ and $v_n = 0$. We obtain them for fixed wheels by setting $d = 0$ and $\beta = 0$ in system 1.5:

$$\begin{bmatrix} v_t \\ v_n \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & L \cdot \sin\gamma \\ -\sin\psi & \cos\psi & L \cdot \cos\gamma \end{bmatrix} {}^m\mathbf{\Omega}_0 \cdot \dot{\xi} + \begin{bmatrix} -r \\ 0 \end{bmatrix} \dot{\varphi} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.7}$$

Clearly, it is always possible to satisfy the non slipping constraint $v_t = 0$, whatever the values of the geometric parameters of the wheel, and whatever the twist of the robot platform[5].

The non skidding constraint $v_n = 0$ means that the velocity of point B of the wheel (wheel center point) is perpendicular to the fixed spin axis y_i of the wheel. Now consider the robot platform: as any solid in a planar motion, its instantaneous motion is a rotation around its ICR (Instantaneous Center of Rotation). A fundamental property of the ICR is that it is located at the intersection of the normals to all velocity vectors of the solid. In other words, the velocity of any point of the solid is perpendicular to the line defined by the point and the ICR. The center of the wheel B is fixed with respect to the platform, hence it belongs to the platform as a solid. Since the velocity of point B is perpendicular to the wheel axle, then the ICR must be located on the wheel axle.

This has important consequences:

- For a configuration with more than two fixed wheels whose axles do not intersect at a common point, as on figure 1.11, case (a), no motion will be compatible with the constraints (*i.e.* any motion will cause at least one wheel to slip or skid).
- If a robot has two or more fixed wheels with parallel axles, as on figure 1.11, case (b), then the only motion compatible with the constraints is a straight line motion perpendicular to the wheel axles.
- A configuration with two or more fixed wheels whose axles intersect at a common point I, as on figure 1.11, case (c), only allows rotation around I without skidding.
- The only kinematically interesting configuration of multiple fixed wheels for a mobile robot is when they have a common axle, as on figure 1.11, case (d). All of them generate identical constraints. Combinations of translations perpendicular to the common axle and rotation around any point of the axle are compatible with the constraint. When there is a rotation component, each wheel spins at a different speed.

The reader may be thinking: *"Is this guy nuts? He takes us along the (not quite fun) process of writing a bunch of equations, just to forget about them and deduce everything*

[5]Of course, in practice any joint has speed limitations, particularly if it is actuated. They are not taken into account here.

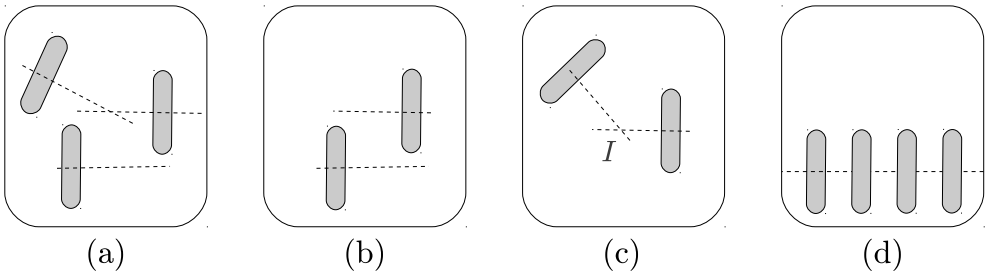


Figure 1.11: Several configurations of fixed wheels, of which the rightmost only is kinematically interesting for a mobile robot.



Figure 1.12: The Pioneer P3 AT skid steering robot.

from physical considerations!" But before throwing the first stone, read the following, as there *are* reasons to write the equations.

By extracting $\dot{\varphi}$ from equation 1.7, you can calculate:

- The rotation speed at which an unactuated wheel will spin (as long as friction at the contact point is sufficient).
- The desired speed at which an actuated fixed wheel should be set when the desired motion is $\dot{\xi}$.

The second case is fundamental to control a robot. Of course, we do not have the full picture yet. The reader may wonder what happens if the motion is not compatible with the constraints, or how to calculate motions that are. This will become clear later in the course.

It is important to know that, in practice, there are robots with fixed wheels on two parallel axles, like the Pioneer P3 AT (AT stands for All Terrain) shown on figure 1.12. Both left wheels are driven at a common speed and both right wheels at a common, independent speed. Fixed wheels are extremely rugged, and this is the reason for choosing to use this configuration. Such robots are called *skid steering robots* because any rotation causes skidding, just as with tracked vehicles. Kinematic modelling is of little use here. A common assumption is that the robot is equivalent to two wheels on a central axle. It sort of works if there is a good symmetry in the repartition of forces on the four wheels, but can give very bad results when it is not the case, particularly on uneven terrain. For a more complete attempt at modeling such robots, refer to [Kozlowski and Pazderski, 1999].

1.4.3 Constraints for steerable wheels

We obtain them by setting $d = 0$ and $\beta = 0$ in system 1.5:

$$\begin{bmatrix} v_t \\ v_n \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & L\cdot\sin\beta \\ -\sin\psi & \cos\psi & L\cdot\cos\beta \end{bmatrix} {}^m\mathbf{\Omega}_0 \cdot \dot{\xi} + \begin{bmatrix} -r \\ 0 \end{bmatrix} \dot{\varphi} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.8}$$

For the satisfaction of the non slipping constraint $v_t = 0$, the situation is similar to the fixed wheel: it is always possible to set $\dot{\varphi}$ so as to satisfy it.

As to the non skidding constraint $v_n = 0$, the situation is similar to a fixed wheel, except the constraint depends on the orientation β of the wheel: the velocity of B must lie in the plane of the wheel for the constraint to be satisfied.

Motorization will be the subject of the last part of the kinematic modeling course, but we can already point out an important characteristic of steerable wheels: their orientation axis *must be motorized*. Indeed, whatever the motion of the robot platform, the spin axis of the wheel will have to be oriented in such a way that it passes through the desired ICR. This will not happen naturally. Now if this explanation is not clear to you, imagine you want to make the robot move in a certain (imposed) way. If the current orientation of the steerable wheel is not correct for that particular motion (*i.e.* the spin axis does not pass through the desired ICR), you first need to orient the wheel (the platform will not move) before you can perform the motion. With a motionless platform, nothing but an actuator can orient the wheel. Later in the course, we will give a more theoretical explanation for the mathematically inclined.

From considerations about the ICR, it is easy to deduce that:

- A robot can have no more than two independent steerable wheels. Indeed, the first two define the ICR (possibly at infinity if their spin axes are parallel); any other steerable wheel will have to be controlled in such a way that its spin axis goes through the ICR as defined by the first two.
- A steerable wheel cannot have its point B on the axle of a fixed wheel. Its orientation would have to be fixed perpendicular to the axle or else the ICR would be fixed at B. In which case, why use a steerable wheel?

1.4.4 Constraints for castor wheels

We obtain them by setting $\gamma = 0$ in system 1.5:

$$\begin{bmatrix} v_t \\ v_n \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & L\cdot\sin\beta \\ -\sin\psi & \cos\psi & d + L\cdot\cos\beta \end{bmatrix} {}^m\mathbf{\Omega}_0 \cdot \dot{\xi} + \begin{bmatrix} 0 \\ d \end{bmatrix} \dot{\beta} + \begin{bmatrix} -r \\ 0 \end{bmatrix} \dot{\varphi} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.9}$$

For the satisfaction of the non slipping constraint $v_t = 0$, the situation is the same as for fixed wheels and steerable wheels. The peculiarity of castor wheels is that we have a similar result for the non skidding constraint $v_n = 0$. Indeed, whatever the values of the geometric parameters and of the twist of the platform, $\dot{\beta}$ can adapt in such a way that $d.\dot{\beta}$ compensates the other terms. So whatever its current orientation, the castor wheel can adapt to any platform motion without violating pure rolling constraints.

This is the reason why castor wheels are so commonly used to make various kinds of carts, particularly shopping carts. If the shopping cart has only castor wheels, then whatever the current orientation of the wheels, you can move the cart in any direction with any rotation speed.

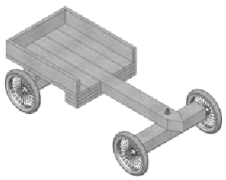


Figure 1.13: A soap box. The two front wheels are not oriented independently.



Figure 1.14: The AIV (Automated Industrial Vehicle) Revolution by Gaussin. The kinematic structure is made of two pairs of rigidly linked general wheels. The AIV is a fully electrical autonomous vehicle used to transport containers in port terminals. It weighs approximately 20 tons and can transport over 60 tons of payload.

1.4.5 Constraints for other wheel assemblies

Now that we have seen what happens in the simpler cases, let us take another look at the general case of system 1.5, assuming that $d\,cos\gamma$ is not equal to zero. Whatever $\dot{\xi}$ and β, $\dot{\beta}$ can be set in such a way that the non skidding constraint $v_n = 0$ is satisfied. It is then easy to determine $\dot{\varphi}$ to satisfy the non slipping constraint $v_t = 0$.

But this will not work when $\gamma = \pi/2$ (or $-\pi/2$), when the equations become:

$$\begin{bmatrix} v_t \\ v_n \end{bmatrix} = \begin{bmatrix} \cos\psi & \sin\psi & d + L\cdot\cos\beta \\ -\sin\psi & \cos\psi & -L\cdot\sin\beta \end{bmatrix} {}^m\mathbf{\Omega}_0 \cdot \dot{\xi} + \begin{bmatrix} d \\ 0 \end{bmatrix} \dot{\beta} + \begin{bmatrix} -r \\ 0 \end{bmatrix} \dot{\varphi} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \qquad (1.10)$$

Here again, satisfying $v_t = 0$ will not be a problem. Satisfying $v_n = 0$ in turn requires $\dot{\xi}$ to be compatible with the current value of β. So the general wheel assembly with $\gamma = \pm\pi/2$ will limit the mobility of the robot in a way similar to the steerable wheel.

This kind of wheels are often used in symmetrical, rigidly linked pairs, like in the soap box (figure 1.13) or the AIV (figure 1.14). It was also found in some horse-powered carriages and early automobiles. The solution was rapidly abandoned on automobiles, first because the vehicle is especially unstable in curves and also because the axis of orientation common to the two general wheels is subject to high mechanical stress and easily damaged. When used in a rigidly linked pair, the two wheels are kinematically equivalent to a single steerable wheel located halfway between the two general wheel assemblies.

Such wheels are rarely used in robotics, but figure 1.15 shows an example, the MPO-700 of the Neobotix company (left) and the type of wheels it uses (right).



Figure 1.15: The Neobotix MPO-700 robot and the type of wheels it uses (source http://www.neobotix-robots.com)

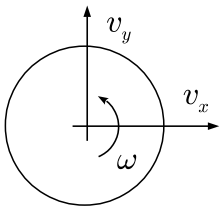


Figure 1.16: The three degrees of freedom of a solid moving in a plane.

1.5 Definition of the Degree of mobility

Figure 1.16 shows the three possible degrees of freedom of a solid (in our case the robot platform) moving in a plane. For a robot, if the wheel configuration allows *independently* generating twist components v_x, v_y and ω *while maintaining pure rolling*, then the *degree of mobility* δ_m of the robot is three, which is the maximum for planar motion. Such a robot is said to be *omnidirectional*.

Beware. *Keep in mind the precise meaning of omnidirectional: any twist should be possible* whatever the current wheel configuration*. If moving in a certain direction and/or with a certain rotation speed requires prior orientation of one or more wheels, then the robot is* not *omnidirectional in the sense defined here.*

In the cases where not all posture speeds are possible, δ_m will be less than three. If $\delta_m = 0$, the wheel configuration is of no practical interest. But $\delta_m = 2$ and $\delta_m = 1$ are both frequent in mobile robots.

We are now going to formalize the notion of degree of mobility and give a way to calculate it.

Beware. *From now on, only* conventional wheels *are taken into account,* i.e. *fixed, steerable and castor wheels.*

Let us precise or recall a few notations:

- Indices "f", "s" and "c" are respectively usef for fixed, steerable and castor wheels.
- The total number of wheels is $N_t = N_f + N_s + N_c$.
- The orientation angles of steerable wheels are concatenated in column vector β_s (dimension N_s), the orientation angles of castor wheels in vector β_c (dimension N_c), in the order of numbering in the parameter table 1.1. The concatenation of β_s and

β_c forms the column vector of all wheel orientations $\beta = [\beta_s^T\ \beta_c^T]^T$, the dimension of which is $N_s + N_c$. Vector β is a function of time $\beta(t)$.

- Likewise, we define the vector of the angular spin of all wheels as: $\varphi = [\varphi_f^T\ \varphi_s^T\ \varphi_c^T]^T$, which is also a function of time $\varphi(t)$.
- The notation of the posture is maintained as in equation 1.1.
- The concatenation of the posture, the orientation angles of the wheels and the spin angles of the wheels is called the *configuration vector* and denoted q. The configuration vector completely specifies the location in space of all the solids which compose the robot: platform and wheel assemblies.

Globally, we have:

$$\xi = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_s \\ \beta_c \end{bmatrix} \quad \varphi = \begin{bmatrix} \varphi_f \\ \varphi_s \\ \varphi_c \end{bmatrix} \quad q = \begin{bmatrix} \xi \\ \beta_s \\ \beta_c \\ \varphi \end{bmatrix} \tag{1.11}$$

1.5.1 Constraints for all wheels in matrix form

The set of N_t non slipping constraints $v_t = 0$ can be put in matrix form:

$$\mathbf{J}_1(\beta_s, \beta_c) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} + \mathbf{J}_2 \cdot \dot{\varphi} = 0 \tag{1.12}$$

Equation 1.12 takes into account the fact that there is no term in $\dot{\beta}$ in non slipping constraints for conventional wheels, and that there is no angle β for fixed wheels.

As to the set of non skidding constraints $v_n = 0$, their maxtrix form is written as:

$$\mathbf{C}_1(\beta_s, \beta_c) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} + \mathbf{C}_2 \cdot \dot{\beta}_c = 0 \tag{1.13}$$

Equation 1.13 takes into account the fact that there is no $\dot{\varphi}$ in non skidding constraints for conventional wheels.

The following equations precise the sizes of the various matrices and how they are decomposed in blocks corresponding to the three kinds of wheels.

$$\mathbf{J}_1(\beta_s, \beta_c) = \overbrace{\begin{bmatrix} \mathbf{J}_{1f} \\ \mathbf{J}_{1s}(\beta_s) \\ \mathbf{J}_{1c}(\beta_c) \end{bmatrix}}^{3} \begin{matrix} |N_f \\ |N_s \\ |N_c \end{matrix} \tag{1.14}$$

$$\mathbf{J}_2 = -r\ \mathbf{I}_{N_t * N_t} \tag{1.15}$$

$$\mathbf{C}_1(\beta_s, \beta_c) = \overbrace{\begin{bmatrix} \mathbf{C}_{1f} \\ \mathbf{C}_{1s}(\beta_s) \\ \mathbf{C}_{1c}(\beta_c) \end{bmatrix}}^{3} \begin{matrix} |N_f \\ |N_s \\ |N_c \end{matrix} \tag{1.16}$$

$$\mathbf{C}_2 = \overbrace{\begin{bmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{C}_{2c} \end{bmatrix}}^{N_c} \begin{matrix} |N_f \\ |N_s \\ |N_c \end{matrix} \tag{1.17}$$

$$\mathbf{C}_{2c} = diag\begin{pmatrix} d_1 & \cdots & d_{Nc} \end{pmatrix} \tag{1.18}$$

With no loss of generality, equation 1.18 is written as though castor wheels were numbered from 1.

Beware. *A matrix can be empty, for example* $\mathbf{C}_{2c}$ *when there are no castor wheels. In such a case, writing* $\mathbf{C}_{2c} = 0$ *is* incorrect. *You can write* $\mathbf{C}_{2c} = [\,]$ *or even* $\mathbf{C}_{2c} = \emptyset$*, but* certainly not $\mathbf{C}_{2c} = 0$.

1.5.2 Satisfying non slipping constraints

The set of non slipping constraints $v_t = 0$ has been written in matrix form in equation 1.12. Since $\mathbf{J_2}$ (equation 1.15) is clearly invertible, the constraints can be satisfied by setting:

$$\dot{\varphi} = \frac{1}{r}\left[\mathbf{J}_1(\beta_s, \beta_d) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi}\right] \tag{1.19}$$

The fact that we can determine a proper $\dot{\varphi}$ for *any* $\dot{\xi}$ means that non slipping constraints do not cause loss of mobility for any of the wheel types we consider. Of course, this was fairly evident physically and did not require all the math. We had also pointed this out earlier for each kind of wheel. But the remarks about the use of equation 1.7 also hold here, applied to all conventional wheels: equation 1.19 is useful in simulating and controlling the robot.

1.5.3 Satisfying non skidding constraints for castor wheels

The set of non skidding constraints $v_n = 0$ for castor wheels has the following matrix form:

$$\mathbf{C}_{1c}(\beta_c) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} + \mathbf{C}_{2c} \cdot \dot{\beta}_c = 0 \tag{1.20}$$

Since $\mathbf{C}_{2c}$, defined in equation 1.18, is invertible, the constraints can be satisfied by setting:

$$\dot{\beta}_c = -\mathbf{C}_{2c}^{-1} \cdot \mathbf{C}_{1c}(\beta_c) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} \tag{1.21}$$

with:

$$\mathbf{C}_{2c}^{-1} = diag\begin{pmatrix} \frac{1}{d_1} & \cdots & \frac{1}{d_{Nc}} \end{pmatrix}$$

So the non skidding constraints of castor wheels do not cause loss of mobility. Here again, this conclusion is not the only interest of equation 1.21, as it will be useful in simulating and controlling the robot.

1.5.4 Satisfying non skidding constraints for fixed and steerable wheels

The set of non skidding constraints $v_n = 0$ for fixed and steerable wheels has the following form:

$$\begin{cases} \mathbf{C}_{1f} \cdot {}^{m}\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} = 0 \\ \mathbf{C}_{1s}(\beta_s) \cdot {}^{m}\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} = 0 \end{cases} \tag{1.22}$$

These equations mean that the twist, expressed in R_m $({}^{m}\mathbf{\Omega}_0(\theta)\,\dot{\xi} = {}^{m}\dot{\xi})$ must belong to the kernel of the sub-matrix of $\mathbf{C}_1$ corresponding to the fixed and steerable wheels. System 1.22 can also be mathematically expressed as:

$${}^{m}\mathbf{\Omega}_0(\theta)\,\dot{\xi} \in Ker\left(\mathbf{C}_1^*(\beta_s)\right), \quad with: \ \mathbf{C}_1^*(\beta_s) = \begin{bmatrix} \mathbf{C}_{1f} \\ \mathbf{C}_{1s}(\beta_s) \end{bmatrix} \tag{1.23}$$

By definition, the *degree of mobility* of the robot is the dimension of vector space $Ker\left(\mathbf{C}_1^*(\beta_s)\right)$. Each independent line of $\mathbf{C}_1^*$ lowers the dimension of $Ker\left(\mathbf{C}_1^*(\beta_s)\right)$:

$$\delta_m = dim\left[Ker\left(\mathbf{C}_1^*(\beta_s)\right)\right] = 3 - rank\left(\mathbf{C}_1^*(\beta_s)\right) \leq 3 \tag{1.24}$$

Of course, as soon as the robot has at least one fixed or steerable wheel, the rank of C_1^* will be at least 1, and hence robot mobility will be restricted. To what extent it is restricted depends on how many *independent* constraints $\mathbf{C}_1^*$ contains, which is what the rank expresses. The example of figure 1.11, case d, shows that several wheels may generated identical constraints, equivalent to a single one.

Remark. *The reader may wonder why robot manufacturers use steerable and fixed wheels, at the expense of mobility? The reasons do not lie in kinematics, as we have seen. The problem with castor wheels is that they are not adapted to very high loads, due to the lever arm d, which causes high moments in the joints. There are also other problems, particularly when inverting the direction of motion of a castor wheel, which must then rotate* 180° *around its orientation axis: high forces can be generated, and the direction in which the wheel will turn is unpredictable. Because the load force goes through the contact point with the ground, fixed and steerable wheels are much more rugged.*

Beware. *In an exam, pay attention to the text. If you are only required to calculate* δ_m*, then you just need to calculate* $\mathbf{C}_1^*$ *and you should not calculate* $\mathbf{J}_1$*, nor should you calculate* $\mathbf{C}_{1c}$*, especially since equations for the castor wheels are a bit more complex than for fixed and steerable wheels.*

1.6 The various types of three-wheeled robots

In this section, we are going to see that, given the constraint equations we have established, there are only a limited number of kinematically reasonable three-wheel configurations.

Remark. *The reason for considering three-wheel configurations is that it is the minimum number of wheels for a statically stable robot.*

1.6.1 Degree of steerability

Definition 6 (Degree of steerability)**.** *The* degree of steerability *is defined as* $\delta_s = rank(\mathbf{C}_{1s})$.

It corresponds to the number of steerable wheels which can be independently oriented. By considering the ICR, we already pointed out earlier that:

$$0 \leq \delta_s \leq 2 \tag{1.25}$$

Again, by considering the ICR, we can deduce that, when $\delta_s = 2$, the robot can have no fixed wheel ($N_f = 0$). Furthermore, since a steerable wheel generates a constraint, $\delta_s = 0$ implies that there is no steerable wheel ($N_s = 0$).

In an analogous way, we can define $\delta_f = rank\,(\mathbf{C}_{1f})$.

Since a fixed wheel creates a constraint, δ_f is 0 if and only if $N_f = 0$.

In section 1.4.2, using the notion of ICR we already studied various configurations of fixed wheels. Figure 1.11.a, we pointed out as leaving no mobility. This case corresponds to $\delta_f = 3$: each wheel generates an *independent* constraint. Mathematically, $Ker\,(\mathbf{C}_1^*) = Ker\,(\mathbf{C}_{1f}) = \left\{\vec{0}\right\}$. The only motion compatible with pure rolling constraints is the null vector.

On figures 1.11.b and 1.11.c, the ICR is fixed, at infinity in case b, at the intersection of the wheel axles in case c, and $\delta_f = 2$.

Exercise 4. *Consider the configuration of three fixed wheels in Figure 1.11.a. Prove by calculation of the rank of* bfC_1^* *that* $Ker\,(\mathbf{C}_1^*) = \left\{\vec{0}\right\}$. *You can choose a particular configuration which simplifies calculations, as long as the axes of the three fixed wheels do not meet at a common point.*

Exercise 5. *Consider the configuration of two fixed wheels in Figure 1.11.c. Prove by calculation of the rank of* bfC_1^* *that* $Ker\,(\mathbf{C}_1^*) = \left\{\vec{0}\right\}$.

Remark. *it might seem strange in exercise 5 to consider a configuration with only two wheels. What about stability? This is of course not a problem. You can always imagine that one castor wheel is added somewhere for stability, and ignored in calculations. Indeed, as they do not restrict mobility, they are not taken into account in* $\mathbf{C}_1^*$ *and can be ignored altogether in the determination of* $Ker\,(\mathbf{C}_1^*)$. *Alternatively, you could add one more fixed wheel whose axle goes through the point of intersection of the first two, as exemplified by some micro-wave oven rotating trays.*

On figure 1.11.d, where several fixed wheels are mounted on the same axle, the wheels do not create independent constraints but a single constraint, and hence $\delta_f = 1$. I shall not detail this case here, as it is detailed in the class exercise which covers the full kinematic modeling of the popular differential drive robot.

Since cases a ($\delta_f = 3$), b ($\delta_f = 2$) and c ($\delta_f = 2$) of figure 1.11 are not kinematically interesting, we can say that *in cases of interest* the following inequations hold:

$$0 \leq \delta_f \leq 1 \tag{1.26}$$

Furthermore, if we want a non zero mobility, it is clear that:

$$0 \leq \delta_s + \delta_f \leq 2 \tag{1.27}$$

By considering the various cases for δ_s, it is also easy to establish the following constraint:

$$2 \leq \delta_m + \delta_s \leq 3 \tag{1.28}$$

Beware. *Always keep in mind that the above inequations hold only* in cases of interest*, in other words* for kinematically reasonable robots.

1.6.2 The five types of mobile robots

Definition 7. *The* type *of a mobile robot is the ordered pair of values* (δ_m, δ_s).

The value of δ_f is a consequence of the values of δ_m and δ_s and is not stated in the type. Given the constraints expressed by equations 1.25 to 1.28, the possibilities are:

Table 1.2: The different types of mobile robots

δ_m	3	2	2	1	1
δ_s	0	0	1	2	1

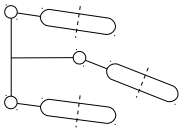
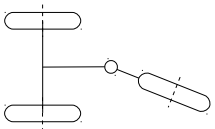
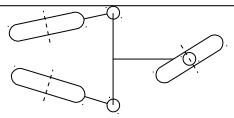
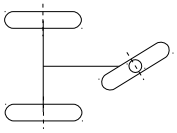
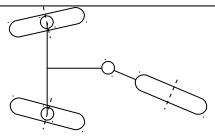
Exercise 6. *The details of derivation of the informations of table 1.2 have not been given. Make sure you can rigorously derive them.*

Exercise 7. *If you have correctly solved exercise 6, you now know that the information* (δ_m, δ_s) *is equivalent to* $(\delta_m, \delta_s, \delta_f)$*. Now for each type* (δ_m, δ_s)*, and assuming the robot has exactly three wheels, determine the wheel configuration of the robot. The results are actually given below: the exercise consists in justifying them in a clear and concise way.*

1.6.3 The five wheel configurations

Table 1.6.3 lists the wheel configurations corresponding to the five types of mobile robots.

Table 1.3: The five types of robots with conventional wheels.

Type	Inform.	Schematics	Remarks
$(3,0)$	$N_f = 0$ $N_s = 0$ $N_c = 3$		It is called *omnidirectional* robot. It can move in any direction with any rotation speed, regardless of the current wheel orientations.
$(2,0)$	$N_f = 2$ $N_s = 0$ $N_c = 1$		Called *differential* robot because rotation is obtained by difference of rotation speed between left and right wheel. Also called *unicycle* robot because it is kinematically equivalent to a single wheel located midway between the left and right wheel.
$(2,1)$	$N_f = 0$ $N_s = 1$ $N_c = 2$		$\dot{\xi}$ belongs to a $2D$ space parameterized by the angle β of the steerable wheel.
$(1,1)$	$N_f = 2$ $N_s = 1$ $N_c = 0$		$\dot{\xi}$ belongs to a 1D space parameterized by the angle β of the steerable wheel. This is the standard model for a car, with both steerable wheels represented by an equivalent central steerable wheel.
$(1,2)$	$N_f = 0$ $N_s = 2$ $N_c = 1$		$\dot{\xi}$ belongs to a 1D space parameterized by the angles β of the steerable wheels.

Chapter 2

Kinematic models and motorization

2.1 What are our goals ?

A standard equation for a non linear system is of the form $\dot{z} = f(z, u)$. In such an equation, u is called the *input*, and z the *state*. Typically, u is a vector which contains the variables that can be used to act on the system to obtain a desired behavior.

Our aim is to obtain such an equation, with the problem that we don't know yet what variables u might contain. We would like u to contain the speed of the motors, or at least some combination of those, but we haven't yet solved the problem of which joints to motorize.

Since we are interested in controlling the robot platform, it is clear that the state z will contain the posture (x, y, θ) of the robot.

In the sequel, we are going to present the tools necessary to:

- Determine the function f.
- Determine a proper vector u, the dimension of which will vary depending on the type of robot.
- Define which joints to motorize to produce the vector u.

2.2 Posture kinematic model

In the posture kinematic model, we are only interested in expressing the twist of the platform in R_0, as a function of the speed (and possibly orientation) of the wheels.

In chapter 1 we have established that the only constraints which limit the mobility of the robot are the non skidding constraints of fixed and steerable wheels, constraints which are concatenated into matrix C_1^*. The constraints state that the twist of the platform $\dot{\xi}$ must belong to the kernel of C_1^*. This was expressed in equation 1.23, which we recall here for convenience.

$$ {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} \in Ker\left(\mathbf{C}_1^*(\beta_s)\right) , \quad with: \ \mathbf{C}_1^*(\beta_s) = \begin{bmatrix} \mathbf{C}_{1f} \\ \mathbf{C}_{1s}(\beta_s) \end{bmatrix} \tag{2.1} $$

The twist of the platform expressed in the robot frame ${}^m\dot{\xi} = {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi}$ can be decomposed over a base of $Ker\left(\mathbf{C}_1^*(\beta_s)\right)$:

$$ {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} \;=\; \Sigma(\beta_s) \cdot u_m \tag{2.2} $$

By definition of δ_m, matrix $\Sigma(\beta_s)$ has δ_m independent columns forming a base of $Ker\left(\mathbf{C}_1^*(\beta_s)\right)$, which is a vector space of dimension δ_m, and u_m is a vector of δ_m components. As indicated in equation 2.2, Σ depends on the orientation β_s of the steerable wheels of the robot. If the robot has no steerable wheels, Σ is a constant matrix.

Multiplying equation 2.2 left by $\mathbf{{}^0\Omega_m}$, we obtain:

$$ {}^0\dot{\xi} \;=\; {}^0\mathbf{\Omega}_m \cdot (\theta) \cdot \Sigma(\beta_s) \cdot u_m \tag{2.3} $$

If the robot has no steerable wheel, Σ is a constant and the equation simply writes:

$$ {}^0\dot{\xi} \;=\; {}^0\mathbf{\Omega}_m(\theta) \cdot \Sigma \cdot u_m \tag{2.4} $$

The equation has the desired form $\dot{z} = f(x, u)$, with $z = \xi$, and constitutes the posture kinematic model of the robot in the case the robot has no steerable wheel. In fact, equation 2.4 obeys a more restrictive form: $\dot{z} = B(z) \cdot u$. We will see that this is the case for the kinematic model of all mobile robots.

If the robot has steerable wheels, the posture kinematic model can be written:

$$ \left\{ \begin{array}{l} {}^0\dot{\xi} \;=\; {}^0\mathbf{\Omega}_m(\theta) \cdot \Sigma\left(\beta_s\right) \cdot u_m \\ \dot{\beta} = u_s \end{array} \right. \tag{2.5} $$

Again, the equation corresponds to the form $\dot{z} = B(z) \cdot u$, with:

$$ \mathbf{B}(z) = \left[\begin{array}{cc} {}^0\mathbf{\Omega}_m(\theta) \cdot \Sigma(\beta_s) & 0 \\ 0 & I \end{array} \right],\; u = \left[\begin{array}{c} u_m \\ u_s \end{array} \right],\; z = \left[\begin{array}{c} \xi \\ \beta_s \end{array} \right] \tag{2.6} $$

Equation 2.5 is the state space form we were looking for, in the case the robot has steerable wheels.

As shown in equation 2.6, the state vector must include vector β_s since the instantaneous kinematic behavior, the twist $\dot{\xi}$ of the robot, depends on the angle of the steerable wheel(s). It should also be noted that the speed of orientation of all steerable wheels will be inputs of the system (vector u_s). It was already physically clear that the orientation joint of steerable wheels would have to be motorized, a fact we pointed out in section 1.4.3, and it is now confirmed by the form of the posture kinematic model.

2.3 Configuration kinematic model

In section 1.5, we have defined the configuration vector of the robot at equation 1.11. We recall here that the configuration vector q is defined as:

$$ q = \left[\xi^T, \beta_s^T, \beta_c^T, \varphi^T\right]^T $$

The *configuration kinematic model* is the expression of $\dot{q}$ as a function of the inputs u_s (if any) and u_m. Of course, the expressions of $\dot{\xi}$ and of $\dot{\beta}_s$ are a part of the configuration kinematic model: the posture kinematic model is a subset of the equations which define the configuration kinematic model. So some of the equations we need in the current section have just been established in the previous one.

We are now left with establishing the expression of $\dot{\beta}_c$ for castor wheels (remember we are considering only conventional wheels) and the expression of $\dot{\varphi}$. Actually, we have already written these equations, respectively in section 1.5.3, equation 1.21 and in 1.5.2, equation 1.19. We recall them here for convenience:

$$\begin{aligned}\dot{\beta}_c &= -\mathbf{C}_{2c}^{-1} \cdot \mathbf{C}_{1c}(\beta_c) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi} \\ \dot{\varphi} &= \frac{1}{r}\left[\mathbf{J}_1(\beta_s, \beta_c) \cdot {}^m\mathbf{\Omega}_0(\theta) \cdot \dot{\xi}\right]\end{aligned}$$

Now we put together, in a block matrix form, the expressions of $\dot{\xi}$, $\dot{\beta}_s$, $\dot{\beta}_c$ and $\dot{\varphi}$, and get the configuration kinematic model. By setting:

$$\begin{cases} \mathbf{D}\left(\beta_c\right) = -\mathbf{C}_{2c}^{-1} \cdot \mathbf{C}_{1c}(\beta_c) \\ \mathbf{E}\left(\beta_s, \beta_c\right) = -\mathbf{J}_2^{-1} \cdot \mathbf{J}_1(\beta_s, \beta_c) \end{cases} \tag{2.7}$$

And remembering from equation 2.2 that ${}^m\mathbf{\Omega}_0(\theta)\dot{\xi} = \Sigma\left(\beta_s\right) u_m$, we obtain:

$$\dot{q} = \mathbf{S}(q)u, \;\; where\; \mathbf{S}(q) = \begin{bmatrix} {}^0\mathbf{\Omega}_m(\theta) \cdot \Sigma(\beta_s) & 0 \\ 0 & \mathbf{I} \\ \mathbf{D}(\beta_c) \cdot \Sigma(\beta_s) & 0 \\ \mathbf{E}(\beta_s, \beta_c) \cdot \Sigma(\beta_s) & 0 \end{bmatrix}, \; and\; u = \begin{bmatrix} u_m \\ u_s \end{bmatrix} \tag{2.8}$$

This last equation completes the configuration kinematic model.

2.4 Motorization

In the previous sections, we have defined the inputs to the configuration kinematic model u_s and u_m. We know that each steerable wheel requires an orientation motor. Now we have to figure out how to produce the vector u_m, *i.e.* which joints to motorize in order to move the platform in accordance to the constraints. We will need at least δ_m motors to do that, since δ_m is the dimension of the vector space of allowable motions. Of course, we also need a relation between the corresponding motor speeds and u_m. This relation should be a two-way relation: we must be able to calculate the motor speeds from a desired motion u_m, and vice versa.

In fact, everything we need is in the configuration kinematic model:

$$\begin{bmatrix} \dot{\xi} \\ \dot{\beta}_s \\ \dot{\beta}_c \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} {}^0\mathbf{\Omega}_m(\theta) \cdot \Sigma(\beta_s) & 0 \\ 0 & \mathbf{I} \\ \mathbf{D}(\beta_c) \cdot \Sigma(\beta_s) & 0 \\ \mathbf{E}(\beta_s, \beta_c) \cdot \Sigma(\beta_s) & 0 \end{bmatrix} \begin{bmatrix} u_m \\ u_s \end{bmatrix}$$

From these equations, we extract the lines corresponding to joints which can potentially be equipped with a motor to produce u_m. This includes the spin joint of all wheels, plus the orientation joint of castor wheels. We can discard the orientation joints of steerable wheels, which we know will be motorized but are used to produce u_s, not u_m. We obtain the following system of equations:

$$\begin{bmatrix} \dot{\beta}_c \\ \dot{\varphi} \end{bmatrix} = \begin{bmatrix} \mathbf{D}(\beta_c) \cdot \Sigma(\beta_s) \\ \mathbf{E}(\beta_s, \beta_c) \cdot \Sigma(\beta_s) \end{bmatrix} u_m = \mathbf{F}\left(\beta_s, \beta_c\right) u_m \tag{2.9}$$

System 2.9 is over-determined. The principle is to select δ_m independent lines of $\mathbf{F}(\beta_\mathbf{s}, \beta_\mathbf{c})$, in which case:

- The corresponding sub-system can be inverted.
- The inverse sub-system determines u_m as a function of the joint speeds corresponding to the selected lines.
- Equipping the corresponding joints with motors solves the motorization of the robot.

Of course, since the system of equations is over-determined, there may be several solutions to the joint selection problem.

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